

Constrained Risk Prediction

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Introduction

Accurate prediction of disease risk in the presence of competing risks remains a central challenge in clinical epidemiology. We target individualized predictions of the 5-year risk of type 2 diabetes (T2D) and simultaneously the 5-year risk of the competing risk of death. Common estimation methods, such as Fine-Gray regression and cause-specific Cox regression, target only one cause at a time and are not optimized for a given prediction time horizon. We develop a loss-based estimation framework for the simultaneous prediction of multiple competing risks at a prespecified time horizon. Our approach provides a way to estimate a broad class of existing regression models for competing risks such that the resulting models are optimized for individualized risk prediction. Crucially, the framework enables seamless integration of flexible learner libraries and enables structural constraints during estimation. These may include coherence constraints that guarantee that the predicted risks respect the parameter space and domain-informed shape constraints (e.g., monotonicity in age). Estimation is achieved via constrained optimization of a differentiable loss function, such as the Brier score.

The Problem

Assume we want to predict the risk of diabetes and the risk of death at some timepoint τ , i.e. we want to predict the risk of each cause in a competing risks scenario. Denote T as the time between time0 and the date of an event and $D \in \{0, 1, \dots, K\}$ as the cause of the event, where $D = 0$ is censoring. Let $X = (X^1, \dots, X^p)$ be a p -dimensional vector of baseline covariates, which can both be numeric or categorical.

Consider a setting where $D \in \{0, 1, 2\}$ and X is 1-dimensional. Define the cumulative hazard for $D = j$ as

$$\Lambda_j(t|x) = \exp(x\beta_j)\Lambda_{0j}(t)$$

where Λ_{0j} is the baseline cumulative hazard for cause j , which is defined as $\Lambda_{0j}(t) = \int_0^t \lambda_{0j}(s)ds$, where λ_{0j} is the baseline hazard. The baseline hazard is however not estimated when fitting a cause-specific cox model. Now define the survival function using the product integral as

$$S(t|x) = \prod_{s \leq t} (1 - d\Lambda_1(s|x) - d\Lambda_2(s|x)),$$

or using the exponential approximation

$$S(t|x) = \exp(-\Lambda_1(t|x) - \Lambda_2(t|x)).$$

We can then write the cumulative incidence function for cause j , $F_j(t|x) = P(T \leq t, D = j|X)$, as

$$F_j(t|x) = \int_0^t S(s-|x)d\Lambda_j(s|x).$$

The problem I will look at is that current methods of estimation of the cause-specific risks estimates the cause-specific cumulative incidence functions separately. The problem with this is that it can lead to wrong estimations, which can easily be verified as it can happen that for τ and some x^*

$$F_1(\tau|x^*) + F_2(\tau|x^*) > 1. \quad (1)$$

I will in this project develop a method to estimate the risks simultaneously such that the above can't happen. MAYBE ALSO MENTION THE PROPORTIONAL HAZARDS ASSUMPTION, AND THAT BY USING BRIER-SCORE AS ESTIMATOR WE DON'T NEED THAT ASSUMPTION AS WE OPTIMIZE THE PARAMETERS FOR A GIVEN τ , AND DONT OPTIMIZE FOR THE WHOLE TIME PERIOD. Maybe this should also be the new goal instead of the "estimation of all causes simultaneously" goal. i.e. "In this project we aim to develop an estimation method based on the proportional hazards model that do not need the proportional hazards assumption". This is interesting as the proportional hazards model is so popular and being able to still use it while having an assumption less is huge.

Different loss functions

Likelihood function

Asume we have N datapoints and K different causes. The general parametric likelihood function in a competing risks setting can be written as:

$$L = \prod_{i=1}^N \lambda_{j_i}(t_i|x_i)^{d_i} S(t_i|x_i),$$

where d_i is 1 if an event was observed and 0 if i is censored, and j_i is which event occurred for person i . Using that the survival function is a product of the individual cause-specific survival functions, $S(t_i|x_i) = \prod S_j(t_i|x_i)$, we can write the likelihood as:

$$\begin{aligned} L &= \prod_{i=1}^N \lambda_{j_i}(t_i|x_i)^{d_i} \prod_{j=1}^K \exp(-\Lambda_j(t_i|x_i)) \\ &= \prod_{i=1}^N \prod_{j=1}^K \lambda_{j_i}(t_i|x_i)^{d_{ij}} \exp(-\Lambda_j(t_i|x_i)). \end{aligned}$$

This means that the likelihood function is a product of K likelihoods, one for each cause. We can then maximize this likelihood by maximizing each cause-specific likelihood function, as long as the parameters are different for each cause.

Likelihood estimating equations

Assume two causes, only one covariate and constant baseline hazards. I then differentiate the log-likelihood functions with respect to the parameters. The negative log-likelihood function for cause j is then:

$$-\log(L_j) = \sum_{i=1}^N -d_{ij}(\beta_j x_i + \log(\lambda_{0j})) + t_i \exp(\beta_j x_i) \lambda_{0j}$$

and differentiated

$$\frac{d - \log(L_j)}{d\beta_j} = \sum_{i=1}^N -d_{ij} x_i + t_i x_i \exp(\beta_j x_i) \lambda_{0j}$$

$$\frac{d - \log(L_j)}{d\lambda_{0j}} = \sum_{i=1}^N -\frac{d_{ij}}{\lambda_{0j}} + t_i \exp(\beta_j x_i)$$

Brier score

As our target in risk prediction is to minimize the Brier-score, it makes sense to fit the models by choosing the parameters which minimizes the brier-score. By using the Brier-score as our loss function, we also optimize our parameters based on a single timepoint (the prediction horizon), which is of great interest in risk prediction. The model produced by minimizing the Brier-score will also be fully parametric, which will allow us to easily use it outside of online environments like DST.

Assuming we have N datapoints of the form $(\tilde{T}, \tilde{D}, X)$ where $\tilde{T} \in (0, \infty)$ is the observed time (time to first observed event), $\tilde{D} \in \{1, 2, \dots, k\}$ is the observed event and X is a p -dimensional covariate vector. The Brier-Score for predicting cause j at timepoint τ can be written as

$$BS = \frac{1}{N} \sum_{i=1}^N (F_j(\tau|x_i) - o_i(\tau, j))^2,$$

where $o_i(\tau, j)$ is the true outcome of person i , i.e. $o_i(\tau, j) = 1(\tilde{D}_i = j, \tilde{T}_i \leq \tau)$. Assume K different causes and assume no censoring (with censoring just add a 0 state such that we have $K + 1$ states and the setup will be similar to that of the joint survival super learner). We then want to find the parameters that minimize the Brier-score for all causes i.e.

$$\begin{aligned} BSS &= \arg \min_{\beta \in \mathbb{R}, \Lambda_0 \in \mathbb{R}^+} \sum_{j=1}^K \frac{1}{N} \sum_{i=1}^N (F_j(\tau|x_i, \beta, \Lambda_0) - o_i(\tau, j))^2 \\ &= \arg \min_{\beta \in \mathbb{R}, \Lambda_0 \in \mathbb{R}^+} \frac{1}{N} \sum_{j=1}^K \sum_{i=1}^N \left(\int_0^\tau S(s|x_i, \beta, \Lambda_0) d\Lambda_j(s|x_i, \beta_j, \Lambda_{0j}) - o_i(\tau, j) \right)^2, \end{aligned}$$

where β are the beta coefficients for all causes and Λ_0 are the baseline hazards, with j indicating the coefficients for the specific cause. BSS = Brier-score-sum. This is actually just the Brier-score as it was written by Brier in the original paper.

Find estimating equations in Parametric Cox with exponential baseline hazard

Assume we are in a competing risks setting with two causes and one covariate x , and we are interested in predicting the risk of experiencing the events before or at timepoint τ using a cox model and let $\phi = \{\beta_1, \beta_2, \Lambda_{01}, \Lambda_{02}\}$. First we take the simpler example and use the exponential approximation of the survival function and assume a constant baseline hazard. The hazard for cause j will be

$$\lambda_j = \exp(\beta_j x) \lambda_{0j},$$

where λ_{0j} is some constant. We are interested in solving this:

$$BSS = \arg \min_{\beta_1, \beta_2 \in \mathbb{R}, \lambda_{01}, \lambda_{02} \in \mathbb{R}^+} \frac{1}{N} \sum_{j=1}^2 \sum_{i=1}^N (g_j(\phi) - o_i(\tau, j))^2.$$

First I rewrite $g_1(\phi) = \int_0^\tau S(s|x, \beta, \Lambda_0) \lambda_1(s|x, \beta_1, \Lambda_{01}) ds$ from the Brier-score written above (note that g_1 is equal to F_1 for a fixed timepoint τ):

$$\begin{aligned} g_1(\phi) &= \int_0^\tau \exp[-\exp(\beta_1 x) \Lambda_{01}(s-) - \exp(\beta_2 x) \Lambda_{02}(s-)] \exp(\beta_1 x) \lambda_{01} ds \\ &= \exp(\beta_1 x) \lambda_{01} \int_0^\tau \exp[-\exp(\beta_1 x) s^- \lambda_{01} - \exp(\beta_2 x) s^- \lambda_{02}] ds \\ &= \exp(\beta_1 x) \lambda_{01} \frac{\exp[\tau^- (-\exp(\beta_1 x) \lambda_{01} - \exp(\beta_2 x) \lambda_{02})] - 1}{-\exp(\beta_1 x) \lambda_{01} - \exp(\beta_2 x) \lambda_{02}} \\ &= \exp(\beta_1 x) \lambda_{01} \frac{1 - \exp[\tau^- (-\exp(\beta_1 x) \lambda_{01} - \exp(\beta_2 x) \lambda_{02})]}{\exp(\beta_1 x) \lambda_{01} + \exp(\beta_2 x) \lambda_{02}} \end{aligned}$$

As we are in continuous time $\tau^- = \tau$. Then we differentiate g_1 with respect to $\beta_1, \beta_2, \lambda_{01}$ and λ_{02} (to get the result for g_2 just swap all the 1's and 2's, also the ones on the left side of the equals sign):

$$\begin{aligned} \frac{\partial g_1(\phi)}{\partial \beta_1} &= \frac{x \lambda_{01} e^{x\beta_1}}{(\lambda_{01} e^{x\beta_1} + \lambda_{02} e^{x\beta_2})^2} \\ &\quad \cdot \left[\lambda_{02} e^{x\beta_2} \left(1 - e^{-\tau(\lambda_{01} e^{x\beta_1} + \lambda_{02} e^{x\beta_2})} \right) + \tau \lambda_{01} e^{x\beta_1} (\lambda_{01} e^{x\beta_1} + \lambda_{02} e^{x\beta_2}) e^{-\tau(\lambda_{01} e^{x\beta_1} + \lambda_{02} e^{x\beta_2})} \right] \\ \frac{\partial g_1(\phi)}{\partial \beta_2} &= \frac{x e^{\beta_1 x} \lambda_{01} e^{\beta_2 x} \lambda_{02}}{(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})^2} \left[(1 + \tau(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})) e^{-(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})\tau} - 1 \right] \end{aligned}$$

$$\frac{\partial g_1(\phi)}{\partial \lambda_{01}} = \frac{e^{\beta_1 x}}{(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})^2} \cdot \left[e^{\beta_2 x} \lambda_{02} \left(1 - e^{-\tau(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})} \right) + \tau e^{\beta_1 x} \lambda_{01} (e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02}) e^{-\tau(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})} \right]$$

$$\frac{\partial g_1(\phi)}{\partial \lambda_{02}} = \frac{e^{\beta_1 x} \lambda_{01} e^{\beta_2 x}}{(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})^2} \left[(1 + \tau(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})) e^{-(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})\tau} - 1 \right]$$

Using this we can now differentiate the Brier-Score (BS) with respect to the betas to get the estimating equations:

$$\begin{aligned} \frac{\partial BSS(\phi)}{\partial \beta_1} &= \frac{1}{N} \sum_{j=1}^K \sum_{i=1}^N \frac{dg_j(\phi)}{\beta_1} 2(g_j(\phi) - o_i(t, j)) \\ &= 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial BSS(\phi)}{\partial \beta_2} &= \frac{1}{N} \sum_{j=1}^K \sum_{i=1}^N \frac{dg_j(\phi)}{\beta_2} 2(g_j(\phi) - o_i(t, j)) \\ &= 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial BSS(\phi)}{\partial \lambda_{01}} &= \frac{1}{N} \sum_{j=1}^K \sum_{i=1}^N \frac{dg_j(\phi)}{\lambda_{01}} 2(g_j(\phi) - o_i(t, j)) \\ &= 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial BSS(\phi)}{\partial \lambda_{02}} &= \frac{1}{N} \sum_{j=1}^K \sum_{i=1}^N \frac{dg_j(\phi)}{\lambda_{02}} 2(g_j(\phi) - o_i(t, j)) \\ &= 0 \end{aligned}$$

Note that the $\frac{1}{N}$ terms can be removed.

Estimation with constraint

Assuming we have N datapoints, the Brier-Score for predicting cause j at timepoint τ can be written as

$$BS_j = \frac{1}{N} \sum_{i=1}^N (F_j(\tau|x_i) - o_i(\tau, j))^2,$$

where $o_i(\tau, j)$ is the true outcome of person i (1 if event j is the observed event and has happened at or before timepoint τ and 0 otherwise). Furthermore

assume only two competing risks (T2D and death). We then want to find the parameters that minimize this score i.e.

$$\arg \min_{\beta \in \mathbb{R}} \sum_{j=0}^K \frac{1}{N} \sum_{i=1}^N (F_j(\tau|x_i, \beta) - o_i(\tau, j))^2.$$

In our setting $K = 2$ where $j = 0$ is the censored state and 1 and 2 are T2D and death. We need to minimize this score under some constraint, for example:

$$\sum_{j=0}^K F_j(\tau|x, \beta) \leq 1 \text{ for all } x,$$

or

$$\frac{d}{d\beta_{age}} F_j(\tau|x_i, \beta) > 0$$

for some j . In the T2D and death example $j \in \{1, 2\}$, as those risks should increase with age. In practise it may be easier to rewrite the minimization problem to:

$$\arg \min_{\beta \in \mathbb{R}} \sum_{j=0}^K \frac{1}{N} \sum_{i=1}^N (F_j(\tau|x_i, \beta) - o_i(\tau, j))^2 + I \left(\frac{d}{d\beta_{age}} F_j(\tau|x_i, \beta) \leq 0 \right),$$

i.e. add the constraint to the loss function as a new term which will be 0 if the constraint holds and 1 otherwise which is large as we are using the Brier-score. We should probably multiply the indicator function with $K + 1$, as we sum $K + 1$ Brier-scores.

If we want methods like Newton-Raphson to work we probably need to use a different method as above, as the loss-function need to be smooth. Say we have the constraint $g(\beta) \leq 0$ then we can use the loss function:

$$\arg \min_{\beta \in \mathbb{R}} \sum_{j=0}^K \frac{1}{N} \sum_{i=1}^N (F_j(\tau|x_i, \beta) - o_i(\tau, j))^2 + \lambda \max(0, g(\beta))$$

where λ is some penalization parameter.

Censoring

The BSS handles censoring well as is, and outperforms everything in the simulation study. Why?

Simulation study

We want to test our method in four different setups: first where the model is correctly specified, second where the model is not correctly specified but

still proportional hazards and one where the proportional hazards assumption doesn't hold. This is done by simulating data from parametric Cox models, with different baseline hazards: first it will be exponentially distributed, second it will be Weibull distributed, and lastly it will be Weibull where the shape parameter will depend on covariates. This will be done both with and without censoring, where the censoring process is Cox-weibull. In all the simulations the models will be fitted using 70 percent of the data, and evaluated by calculating the BSS on the 30 percent of data not used for training.

We test our estimation method against the exponential-Cox model estimated using maximum likelihood, the logistic regression model and the standard Cox proportional hazards model.

Correctly specified model (Sim-setup 1)

In this setup we are interested in two things: See if our method finds the true parameters and compare model performance between BSE and other models. The results of these simulation studies can be found in the `simulation_study` folder in the targets: `doesBSfindcorrectpars`, `simulation_coxexp`, `simulation_coxexp_cens`.

Brier-score estimation does indeed find the data-generating parameters, but can fail if the starting parameters are chosen poorly. When data contains no censoring all the methods perform approximately the same, with the Oracle estimator performing best and the BSE performing worst. The models also perform almost equal when about 30 percent of the data is right censored, but here the BSE performs best and the logistic model performs worst.

Not correctly specified (Sim-setup 2)

All the methods perform approximately the same when there is no censoring in the data. When introducing censoring the BSE outperforms the rest.

Non-proportional hazards model (Sim-setup 3)

When data violates the proportional hazards assumption the BSE outperforms all the other methods, both when data contains censoring and when no censoring is occurring.

Choosing starting parameters

The optimal way to choose starting parameters for the optimizer would be to try many different values for each parameter and choose the best outcome. However this is computationally heavy and time consuming. A good way to choose starting parameters I have found is to use the parameters from a fitted Cox model. Note that we can never be sure if we have found the true minimum of the BSS.

"Real" data example

Here we test our method on synthetic data that mimics the data used for the Steno1-risk-engine (USE DATA FROM DP-NEXT INSTEAD), and show the performance against the Cox-proportional hazards model.